

FFGFT — Epistemological Position

4D Description, Natural Units, and Self-Referentiality

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This document summarises the epistemological position of the Fundamental Fractal Geometric Field Theory (FFGFT). It addresses: (1) the status of the 4D description (fractal, open, recursive), (2) the equivalence of all descriptions in natural units, (3) the structure of the vacuum and the T-field, (4) the FFGFT position on black holes, and (5) the unavoidable self-referentiality of any inside description. A closing remark draws the parallel to the Logos concept of the Gospel of John.

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1. The 4D Description Is Fractal, Open, and Recursive

FFGFT works with the 4D identification torus structure T^4 — but this description is not a closed, deterministic system.

Fractal: The same geometric structure (ξ -ladder, winding numbers) appears at different scales. The fractal dimension $D_f = 2.999867$ is extremely close to 3 — the deviation is subtle but physically consequential.

Open: The first step — why T^4 , why $\xi = 1.333 \times 10^{-4}$ — remains explicitly open. FFGFT makes no ontological completeness claim.

Recursive: The ξ -ladder is a recursive structure with a natural cut-off at $N \approx 8-9$, where the Z_3^3 topological structure and Fibonacci convergence coincide.

2. The 4D Description Does Not Apply to the Totality

FFGFT does *not* claim that the entire domain of existence is described by 4D and nothing further exists beyond it. The 4D description is a framework for what is physically meaningful within its bounds — not a total claim about existence.

FFGFT is not metaphysical absolutism. It is a derivation framework: *given T^4 and ξ , everything else follows*. What lies before T^4 remains open — a deliberate, honest choice.

3. The Lower Bound: 4D Applies to the Absolutely Smallest

$$L_0 = \xi \cdot \square_P \approx 2.2 \times 10^{-39} \text{ m}$$

This is the lower boundary of the 4D framework. Below L_0 , T^4 has no well-defined degrees of freedom. This boundary applies *downward*, not upward. The question of the “total size” of T^4 is structurally ill-posed — torus size is an emergent coherence length, not a fixed parameter.

4. Natural Units: No Privileged Description

$$E \leftrightarrow m \leftrightarrow 1/t \leftrightarrow f$$

In natural units, all fundamental descriptions are equivalent: $E = mc^2$, $E = hf$, $T \cdot m = 1$ (FFGFT’s foundational relation, $T = T$ -field operator), $f = 1/t$. None is privileged. [Pascher 2026]

5. Consequence: Information vs. Geometry

“Information” as a description is a valid projection — but not more fundamental than the geometric description.

Information is distinction. Shannon (1948), Bateson (1972), and the IPI discussions of 2026 converge: without distinction, there is no information. “Pure unrestrained information” without structure is indistinguishable from nothing — it is not a primitive but the absence of everything.

6. The Vacuum Has Structure — No Field Without Structure

6.1. The Vacuum Is Not Empty

In FFGFT, the vacuum is the ξ -field ground state: vacuum energy density as a geometric property of the ξ -field, T-field configuration $T \cdot m = 1$ active even in the ground state, $L_0 = \xi \cdot \ell_P$ as a structural lower bound.

6.1.1 Thermodynamic Equilibrium Changes Nothing

Even in total thermodynamic equilibrium, the field remains structured: field equations still hold, T^4 topology still exists, ξ retains its value.

Equilibrium is not the absence of structure. Equilibrium presupposes a space of structure within which equilibrium is definable. Without field structure, there is no concept of equilibrium.

6.2. FFGFT Is a Field Theory — No Field Without Structure

A field is a mapping from points of a carrier space to values obeying field equations. **No field without structure** — this is a mathematical tautology. “Pure unstructured information” cannot carry a field, cannot build a vacuum, cannot satisfy a field equation.

7. Black Holes — Horizon, Singularity, and Fundamental Information

7.1. Black Holes Are Points in the Universe, Not the Universe

A black hole is a local, isolated region of the universe. The event horizon and the cosmological horizon are **epistemic** boundaries — not ontological statements about what exists beyond them. We can only describe what we see.

7.2. Black Holes Are Not Singular in FFGFT

$L_0 = \xi \cdot \ell_P$ is an absolute lower bound — infinite density is structurally excluded. $T \cdot m = 1$ remains valid even at extreme mass concentrations. Black holes are specific T^4 winding configurations — extreme states within the framework’s validity. [Pascher 2026]

7.3. FFGFT Does Not Lose Fundamental Information

Winding numbers (n_θ , n_ϕ , n_3 , n_4) are *topological invariants* — they cannot disappear. What appears as “information loss” is an **access problem** (epistemic), not an **existence problem** (ontological). [Bekenstein 1973; Hawking 1974]

8. Self-Referentiality — We Always Describe From the Inside

8.1. We Are Part of the System

Our description of FFGFT stands conceptually above FFGFT — but this meta-level is only possible because we are materially structured beings. Neurons, electrochemical signals, conscious processes — all within T^4 .

We are part of the system. We always describe from the inside, never from the outside.

8.2. The Thought Experiment of an Outside View

The attempt to adopt an outside view through the mind is itself a material process within T^4 . Every apparent outside view is an inside view dressed as an outside view.

8.3. The Recursive Situation

Matter → Description → Matter → Description → ...

In FFGFT, this recursion does not terminate — and this is consistent. The framework always returns to $T^4 + \xi$ — not because this is provable from outside, but because it is the primitive posited *from within*.

Gödel Analogy: A sufficiently powerful formal system cannot prove its own consistency from its own axioms [Gödel 1931]. FFGFT cannot derive its own foundation from within — which is why “why T^4 ?” remains explicitly open. This is not a gap; it is structural honesty.

8.4. Consequence for the Postulate of Pure Information

The postulate “pure information before geometry” implicitly assumes an outside view. But this standpoint is itself a material process within T^4 . FFGFT accepts this limit and describes consistently from the inside.

Closing Remark: “In the Beginning Was the Word”

The discussion about the primitive — geometry or information — leads, in principle, consciously or unconsciously, to one of the oldest answers to the same question: “In the beginning was the Word” (John 1:1).

The Logos concept of the Gospel of John is philosophically precise: the Word is structure, order, distinction — exactly what information in the Shannon sense presupposes. The postulate of “pure, unrestrained information” and the Johannine Logos share the same structural problem: both posit a primitive that stands before all structure, and both require someone *outside the system* to speak it.

The Bible resolves this through inspiration — God as an outside perspective made accessible to the human writers. But the writers were human: material beings within the system. The claim of inspiration is itself a claim made from inside about an outside that was never directly accessible.

FFGFT does not take this step. It posits T^4 and ξ as primitive — from within, honestly, without claiming an outside perspective. “Why T^4 ?” remains open. This is structurally more modest than the biblical claim, and more precise.

The difference in one sentence: The Bible claims an outside view it cannot have. FFGFT relinquishes this claim and names the boundary explicitly.

Summary

Claim	FFGFT Position
4D is the only possible description	No — 4D is the framework, ontologically open
4D applies to the totality of existence	No — FFGFT makes no total claim
4D applies to the absolutely smallest	Yes — $L_0 = \xi \cdot \ell_P$ sets the lower bound
Below L_0 nothing more exists (in FFGFT)	Yes — no stable degrees of freedom
Geometry is privileged over E/m/t	No — all projections are equivalent
Unstructured information as primitive	Contradiction — information is distinction
The vacuum is structureless	No — ξ -field ground state, geometrically structured
A field can exist without structure	No — mathematical tautology
Black holes are singular in FFGFT	No — L_0 structurally prevents infinite density
Black holes destroy information	No — topological invariants are preserved
Horizons are ontological boundaries	No — epistemic limits of the observable
We describe FFGFT from outside	No — we are material T^4 processes
The recursion terminates	No — open recursion, consistent with FFGFT
“Why T^4 ?” is answered	No — structural honesty, not a gap
The Bible adopts an outside view	Claim from inside — FFGFT names the boundary instead

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